

## SHANGZHI, China

WMO: 509680

Lat: 45.2061N

Lon: 127.9794E

Elev: 191

StdP: 99.05

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |       | Humidification DP/MCDB and HR |     |       |       |     |       | Coldest Month WS/MCDB |       |     |       | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|-------|-------------------------------|-----|-------|-------|-----|-------|-----------------------|-------|-----|-------|-----------------------|------|-------|
|               |            |       | 99.6%                         |     |       | 99%   |     |       | 0.4%                  |       | 1%  |       |                       |      |       |
|               | 99.6%      | 99%   | DP                            | HR  | MCDB  | DP    | HR  | MCDB  | WS                    | MCDB  | WS  | MCDB  | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)   | (d)                           | (e) | (f)   | (g)   | (h) | (i)   | (j)                   | (k)   | (l) | (m)   | (n)                   | (o)  | (p)   |
| 1             | -30.1      | -27.7 | -32.7                         | 0.2 | -29.9 | -30.5 | 0.2 | -27.3 | 9.0                   | -11.5 | 7.9 | -13.1 | 1.3                   | 220  | 0.514 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 8.5                    | 30.5            | 21.9 | 29.2 | 21.5 | 27.9 | 21.1 | 24.4                | 27.9 | 23.5 | 27.0 | 22.7 | 26.0 | 3.1                  | 220  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  |                   |
| 23.3                            | 18.5 | 26.6 | 22.4 | 17.5 | 25.6 | 21.6 | 16.6 | 24.7 | 75.1          | 28.1 | 71.5 | 27.0 | 68.3 | 26.0 | 27.5              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |    |    |
|-------------------|------|-----|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|----|----|
|                   |      |     | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |    |    |
| 1%                | 2.5% | 5%  | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  | DB | WB |
| (a)               | (b)  | (c) | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |    |    |
| (4) 8.3           | 7.2  | 6.2 | -33.2                      | 32.9 | 3.6                | 1.4 | -35.8                                              | 34.0 | -37.9      | 34.8 | -39.9      | 35.6 | -42.5      | 36.6 |    |    |
| (5)               |      |     | -33.3                      | 25.6 | 3.6                | 1.0 | -35.9                                              | 26.4 | -37.9      | 27.0 | -39.9      | 27.6 | -42.5      | 28.3 |    |    |

## Monthly Climatic Design Conditions

|      |                                                                   | Annual      |       |       |       |       |       |       |       |       |       |       |       |       |       |
|------|-------------------------------------------------------------------|-------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
|      |                                                                   | (d)         | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   |       |       |
| (6)  | Temperatures, Degree-Days and Degree-Hours                        | DBAvg       | 4.1   | -18.3 | -13.4 | -4.0  | 6.4   | 13.8  | 20.1  | 22.8  | 21.1  | 14.8  | 5.9   | -5.5  | -15.5 |
| (7)  |                                                                   | DBStd       | 15.05 | 4.56  | 5.61  | 6.08  | 4.49  | 3.85  | 3.15  | 2.25  | 2.66  | 3.56  | 4.46  | 6.38  | 5.16  |
| (8)  |                                                                   | HDD10.0     | 3507  | 878   | 655   | 434   | 126   | 9     | 0     | 0     | 0     | 5     | 142   | 465   | 792   |
| (9)  |                                                                   | HDD18.3     | 5513  | 1137  | 888   | 692   | 358   | 147   | 17    | 1     | 7     | 114   | 387   | 715   | 1050  |
| (10) |                                                                   | CDD10.0     | 1356  | 0     | 0     | 0     | 18    | 128   | 303   | 398   | 346   | 149   | 13    | 0     | 0     |
| (11) |                                                                   | CDD18.3     | 322   | 0     | 0     | 0     | 0     | 8     | 71    | 141   | 94    | 8     | 0     | 0     | 0     |
| (12) |                                                                   | CDH23.3     | 2331  | 0     | 0     | 0     | 11    | 139   | 569   | 967   | 561   | 82    | 2     | 0     | 0     |
| (13) |                                                                   | CDH26.7     | 569   | 0     | 0     | 0     | 1     | 30    | 164   | 257   | 109   | 8     | 0     | 0     | 0     |
| (14) | Wind                                                              | WSAvg       | 2.6   | 2.6   | 2.7   | 3.0   | 3.2   | 3.0   | 2.6   | 2.3   | 2.2   | 2.2   | 2.5   | 2.7   | 2.6   |
| (15) | Precipitation                                                     | PrecAvg     | 666   | 5     | 7     | 15    | 33    | 50    | 103   | 178   | 148   | 71    | 33    | 16    | 8     |
| (16) |                                                                   | PrecMax     | 1081  | 11    | 19    | 52    | 84    | 86    | 275   | 311   | 389   | 174   | 108   | 46    | 19    |
| (17) |                                                                   | PrecMin     | 450   | 0     | 1     | 1     | 8     | 8     | 22    | 46    | 39    | 18    | 6     | 3     | 2     |
| (18) |                                                                   | PrecStd     | 135   | 3     | 5     | 12    | 22    | 22    | 57    | 72    | 88    | 35    | 23    | 10    | 4     |
| (19) | Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4% DB     | -2.1  | 5.0   | 14.6  | 25.1  | 29.5  | 32.4  | 32.4  | 30.7  | 27.9  | 22.4  | 13.2  | 2.8   |       |
| (20) |                                                                   | 0.4% MCWB   | -3.8  | 0.2   | 7.6   | 12.9  | 16.7  | 20.3  | 23.7  | 23.8  | 19.8  | 13.9  | 7.9   | 0.6   |       |
| (21) |                                                                   | 2% DB       | -5.7  | 2.1   | 10.4  | 21.2  | 27.1  | 30.3  | 30.7  | 29.2  | 25.7  | 18.9  | 9.2   | -1.2  |       |
| (22) |                                                                   | 2% MCWB     | -7.3  | -1.0  | 4.4   | 10.8  | 15.3  | 19.7  | 22.9  | 23.1  | 18.5  | 11.9  | 5.4   | -3.0  |       |
| (23) |                                                                   | 5% DB       | -8.4  | -0.8  | 7.7   | 18.4  | 24.6  | 28.6  | 29.4  | 27.9  | 23.7  | 16.0  | 6.1   | -4.5  |       |
| (24) |                                                                   | 5% MCWB     | -9.7  | -3.4  | 2.6   | 9.6   | 14.8  | 19.4  | 22.7  | 22.3  | 17.2  | 10.3  | 3.2   | -6.1  |       |
| (25) |                                                                   | 10% DB      | -10.8 | -3.7  | 5.2   | 15.4  | 22.1  | 26.8  | 28.1  | 26.5  | 22.0  | 13.6  | 3.4   | -7.3  |       |
| (26) |                                                                   | 10% MCWB    | -11.8 | -5.7  | 1.0   | 8.1   | 13.8  | 19.1  | 22.4  | 21.4  | 15.9  | 9.2   | 0.9   | -8.6  |       |
| (27) | Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4% WB     | -3.6  | 0.9   | 8.2   | 14.0  | 18.7  | 23.0  | 25.9  | 25.4  | 21.5  | 15.1  | 8.6   | 0.9   |       |
| (28) |                                                                   | 0.4% MCDB   | -2.1  | 4.0   | 13.6  | 21.6  | 25.8  | 27.8  | 30.0  | 28.8  | 25.8  | 20.6  | 12.2  | 2.5   |       |
| (29) |                                                                   | 2% WB       | -7.2  | -0.9  | 5.1   | 11.7  | 17.1  | 21.8  | 24.7  | 24.3  | 20.0  | 13.1  | 5.7   | -2.9  |       |
| (30) |                                                                   | 2% MCDB     | -5.8  | 2.2   | 9.5   | 20.0  | 24.7  | 27.0  | 28.2  | 27.7  | 23.4  | 17.2  | 8.7   | -1.3  |       |
| (31) |                                                                   | 5% WB       | -9.7  | -3.3  | 3.0   | 10.0  | 15.7  | 20.9  | 23.8  | 23.3  | 18.6  | 11.3  | 3.4   | -6.0  |       |
| (32) |                                                                   | 5% MCDB     | -8.5  | -0.8  | 7.0   | 17.3  | 22.0  | 26.2  | 27.5  | 26.4  | 22.1  | 14.8  | 5.9   | -4.6  |       |
| (33) |                                                                   | 10% WB      | -11.8 | -5.7  | 1.2   | 8.4   | 14.6  | 20.0  | 23.1  | 22.4  | 17.2  | 9.5   | 1.2   | -8.5  |       |
| (34) |                                                                   | 10% MCDB    | -10.8 | -3.7  | 4.9   | 14.7  | 20.7  | 24.8  | 26.7  | 25.2  | 20.7  | 13.3  | 3.2   | -7.4  |       |
| (35) | Mean Daily Temperature Range                                      | MDBR        | 11.8  | 13.3  | 11.5  | 12.2  | 11.7  | 10.1  | 8.5   | 8.9   | 12.2  | 11.4  | 9.9   | 10.1  |       |
| (36) |                                                                   | 5% DB MCDBR | 14.4  | 15.2  | 13.9  | 17.8  | 16.4  | 13.3  | 10.8  | 10.7  | 14.6  | 15.8  | 12.6  | 12.7  |       |
| (37) |                                                                   | 5% WB MCWBR | 12.8  | 12.0  | 8.6   | 9.2   | 6.8   | 4.9   | 4.5   | 5.0   | 7.5   | 8.9   | 9.0   | 11.1  |       |
| (38) |                                                                   | 5% WB MCDBR | 14.2  | 14.6  | 13.1  | 16.0  | 13.5  | 10.5  | 8.8   | 8.8   | 11.8  | 14.1  | 12.0  | 12.6  |       |
| (39) | Clear-Sky Solar Irradiance                                        | taub        | 0.336 | 0.363 | 0.441 | 0.543 | 0.591 | 0.610 | 0.575 | 0.513 | 0.458 | 0.479 | 0.395 | 0.350 |       |
| (40) |                                                                   | taud        | 2.025 | 1.973 | 1.860 | 1.730 | 1.687 | 1.735 | 1.850 | 1.994 | 2.053 | 1.882 | 1.999 | 2.004 |       |
| (41) |                                                                   | Ebn at Noon | 723   | 782   | 768   | 723   | 705   | 694   | 714   | 744   | 749   | 642   | 640   | 651   |       |
| (42) |                                                                   | Edh at Noon | 105   | 135   | 175   | 218   | 235   | 225   | 198   | 165   | 143   | 148   | 109   | 98    |       |
| (43) | All-Sky Solar Radiation                                           | RadAvg      | 2.15  | 3.25  | 4.27  | 4.75  | 5.28  | 5.54  | 5.02  | 4.71  | 4.22  | 2.94  | 1.95  | 1.69  |       |
| (44) |                                                                   | RadStd      | 0.18  | 0.22  | 0.29  | 0.36  | 0.51  | 0.59  | 0.45  | 0.37  | 0.32  | 0.21  | 0.15  | 0.10  |       |

## Historical Trends

|      | DBAvg                  | Heating |        |       | Cooling |       |     | Degree-Days |         |         |         |
|------|------------------------|---------|--------|-------|---------|-------|-----|-------------|---------|---------|---------|
|      |                        | 99% DB  | 99% DP | 1% DB | 1% WB   | 1% DP |     | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
| (46) | Station Only           | N/A     | N/A    | N/A   | N/A     | N/A   | N/A | N/A         | N/A     | N/A     | N/A     |
| (47) | Regional (4 neighbors) | N/A     | N/A    | N/A   | N/A     | N/A   | N/A | N/A         | N/A     | N/A     | N/A     |

Nomenclature: See separate page